Student's Index Number ……

THE PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT AUTHORITIES

ARUSHA REGIONAL MOCK EXAMINATION

FORM TWO 033 **BIOLOGY**

TIME 2:30HRS **MAY 2026**

INSTRUCTIONS

1. This paper consist of sections A, B and C
2. Answer all questions in sections A,B and C
3. All writing should be in Blue or Black ink, except for drawing that must be drawn in pencil
4. Cellular phones and any unauthorized material are NOT allowed in the examination room
5. Write your examination number on every page of your answer booklet

QUESTION NUMBER	MARKS	EXAMINER'S SIGN
1		
2		
3		
4		
5		
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7		
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9		
10		

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SECTION A (15 Marks)

Answer all questions in this section

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter in the space provided in the table below.

(i) Most of the students have a tendency of bullying Dina because she is 70kg sy weight and 1.5m by height. Which of the following is correct about Dina?
A. She is underweight with 31.1kg/m^2
B. She is obese with 31.1kg/m^2
C. She is normal with 31.1kg/m^2
D. She is very obese with 46.67kg/m^2

(ii) Which of the following is a characteristic of prokaryotic cells?
A. They contain a nucleus
B. They lack membrane bound organelles
C. They have mitochondria
D. They are always multicellular

(iii) The bread mould is an example of
A. Basidiomycota
B. Zygomycota
C. Brayophyta
D. Pteridophyta

(iv) A student is preparing to observe onion cells under a microscope. Which of the following is the correct way to handle the microscope?
A. Carry the microscope with one hands holding the eyepiece
B. Carry the microscope with both hands, one on the arm and the other supporting the base.
C. Tilt the microscope sharply to adjust the view
D. Place the smicroscope near the edge of the table for easy access.

(v) Viruses are classified in any of the major groups of living things because they can exist as living organism or non – living organism. Which of the following is the reason for viruses to be termed as non – living organism?
A. They are infectious when they are inside the host
B. They exist in dormant state when they are outside the host living things
C. They are host specific when they are inside the host
D. They can produce on their own inside the host

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(vi) Which kingdom includes organisms that are, multicellular, autotrophic and have cell walls made of cellulose?

A. Fungi
B. Plantae
C. Animalia
D. Protoctista

(vii) Medical wastes are described as:-

A. Re -usable materials
B. Biohazard wastes
C. Biodegradable
D. Recyclable

(viii) A student is asked to define the term "photosynthesis" which of the following best explains the meaning of this biological term?

A. The breaking down of food to release energy
B. The process by which green plants make food using sunlight
C. The movement of water molecules across a semi – permeable membrane
D. The removal of waste products from the body

(ix) Musa is complaining of having a burning sensation in the esophagus. Which practice would you advice him to alleviate the problem?

A. To engage in physical exercise
B. Ensure good work management
C. Minimize intake of fatty and sugary foods
D. Eating food with enough fiber

(x) A student in Alkheri village observes tap worms in a cows intestine. Which of the following characteristics correctly identifies tape worms as members of the phylum Platyhelminthes?

A. They have a segmented body with jointed appendages
B. They are flat, bilaterally symmetrical and lack a body cavity
C. They posses a hard exoskeleton made of chitin
D. They have a circulatory system with a four chambered heart

Item	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answer										

2. Match the description in List 'A' with corresponding terminologies in List 'B' by writing the correct answer in the table below.

LIST A	LIST B
(i) Biconcave shape, lack nucleus	A. Leucocytes
(ii) Have irregular shape which respond to infections	B. Phagocytes
(iii) Fragments of cells produced in the bone marrow	C. Thrombin
(iv) Fluid tissue consisting of white cells, red cell, platelets and plasma	D. Erythrocytes
(v) Fluid part of the blood	E. Plasma
	F. Capillaries
	G. Thrombocytes
	H. Blood

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Answer

i	ii	iii	iv	v

SECTION 'B' (70 Marks)

3. (a) Define the following terms
(i) Nutrition ……
(ii) Balanced diet ……..
(iii) Nutrient ……..

(b) A form two student spends most of the day sitting in class and when studying at home, with very little physical activity. Explain what type of nutritional requirements such a sedentary student should focus on, and why these requirements differ from those of an active athlete.

(i) ……
(ii) ……
(iii) ……
(iv) ……
(v) ……
(vi) ……..

4. A form one student was presenting his research finding to fellow students. He made a concluding statement that "Scientific research is a never ending process" Give possible reasons for his statement ……..
(i)Identify the step of the scientific method in each of the following situations.
(a) John is not feeling well, he goes to see a medical doctor at a nearby hospital
……
(b) The doctor ask John several questions about how he feels.
……
(c)The doctor orders John's body temperature, blood, and urine samples be measured and diagnosed in the laboratory ……
(d) The laboratory technician diagnoses malaria parasites in John's blood.
……
(e) The doctor confirms that John has malaria and prescribes medicine for him
……

5. (a) Define the following terms
(i) Guttation ……..
(ii) Mass flow ……..

(b) Transpiration is the process which involves different mechanisms so as to ensure that water flows from the roots to the leaves, it's facilitated by different factors. Explain four factors which affect the process.

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(i) ……
……
(ii) ……
……
(iii) ……
……
(iv) ……
……

6. (a)Briefly explain the concepts of the following terms
   (i) Aerobic respiration ……
   (ii) Anaerobic respiration
   ……

   (b)During the practical session from Kinjubi secondary school, most of form two students complained to a teacher due to the lack of enough oxygen within the laboratory room when some chemical bottle is opened, as a biologist student, explain how oxygen and carbon dioxide are transported within the body.

   (i) Transport of oxygen
   ……..
   (ii) Transport of carbon dioxide ……

7. (i) State two differences between bacteria and plant cell

Bacteria cell	Plant cell
(a)	
(b)	

(ii)Draw a well labelled diagram of a bacteria cell

8. (a) Outline four (4) rules observed in binomial nomenclature
   (i) …….
   (ii) ……..
   (iii) …….
   (iv) …….

   (b) Write two advantages of natural classification and two disadvantages of natural Classification.

   Advantages

   (i) …….
   (ii) …….

   Disadvantages

   (i) ……
   (ii) ……

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9. (a) Many digestive disorders is a result from improper eating habits. Explain the consequences of having the following eating habits.
   (i) Eating foods with a lot of sugar …….
   ……
   (ii) Taking meals with no or little fibre ……
   ……
   (iii) Eating food with too much fats, oils or too much spices ……..
   ……..
   (iv) Not drinking adequate amount of water and fluid ……
   ……

   (b) Explain why during digestion the food is ……..

   (i) Alkaline when in the mouth ……..
   ……
   (ii) Acid when in the stomach ……
   ……
   (iii)Alkaline when in the ileum ……..
   ……

SECTION C (15 Marks)

Answer question ten (10)

10. For biological processes to occur in the body, efficient transport of essential material from one place to another is crucial. Use this statement to explain six importance of transportation of materials within the living things.

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